

SLATE: Spectral-Lyapunov Annealed Training for Compressed-Attention Language Models

Rene Claudel Mugenzi Physivitis Ltd, London, United Kingdom rene.mugenzi@physivitis.tech

Version 0.1, 26 May 2026 UK Patent Application: GB2612127.7 (filed 24 May 2026)

Abstract

We introduce SLATE (Spectral-Lyapunov Annealed Training Estimator), a training framework that extends Native Sparse Attention (NSA; Yuan et al., 2025) with three mathematical innovations providing provable training stability guarantees not available in published sparse-attention methods. The three innovations are (i) a spectrally-constrained compression operator G with $\sigma_1(G) \leq 1$, providing 1-Lipschitz behaviour, bounded gradient propagation, and an information-monotonicity guarantee $H(G(X)) \leq H(X)$ via the data processing inequality; (ii) a Fisher-information weighted token selection rule replacing cosine-similarity top-k; and (iii) a Lyapunov-derived annealing schedule with explicit bifurcation guard for the sparse-to-dense transition. We provide validation of the spectral constraint and the combined Innovations 1+3 system at toy scale (1.85M parameters, character-level Shakespeare). The spectral constraint is enforced to within machine precision (max $\sigma_1 = 1.000079$, zero violations across 16 compression layers); information monotonicity is empirically verified by both Gaussian and spectral entropy estimators (zero violations); and quality is preserved within 0.034 validation loss of the unconstrained NSA baseline. Fisher selection and Lyapunov-schedule advantage over heuristics remain theoretically specified with experimental validation deferred to production scale. We position SLATE as a contribution to verified and safety-critical AI training, where empirical stability is insufficient and provable bounds are required.

Keywords: sparse attention, spectral normalisation, Lyapunov stability, language model training, AI safety, verified AI, information monotonicity

1. Introduction

Transformer language models trained with full self-attention incur computational cost scaling quadratically with sequence length, motivating substantial research into sparse and compressed attention mechanisms. Yuan et al. (2025) introduced Native Sparse Attention (NSA), a hardware-aligned natively-trainable method achieving substantial asymptotic speedups at long contexts while preserving benchmark quality. NSA combines coarse-grained compression of key-value pairs with fine-grained selection of compressed summaries via cosine-similarity top-k.

Despite empirical success, published sparse-attention methods suffer from three theoretical deficiencies. First, the compression operator G is learned without operator-norm constraint, meaning G may amplify its inputs, may exhibit unbounded gradients under composition through stacked layers, and may inject information not present in the source tokens. Second, the selection rule for choosing compressed summaries is uniformly cosine similarity, which maximises mutual information between query and key but does not measure the contribution of the selected pair to parameter updates, wasting attention budget on similar-but-uninformative pairs. Third, the schedule controlling the sparse-to-dense transition is chosen heuristically (commonly tanh or cosine), with no guarantee that the transition avoids curvature-degenerate regions of the loss landscape where the

optimiser becomes unstable.

These deficiencies produce observable failure modes in practice: loss spikes at heuristic-schedule transition points, gradient instability through unconstrained compression stacks, and wasted compute on uninformative selected pairs. They are also barriers to deployment in safety-critical, regulated, or formally-verified AI contexts (medical, defence, financial, autonomous systems) where empirical stability is insufficient and provable bounds are required.

We propose SLATE (Spectral-Lyapunov Annealed Training Estimator), a unified training framework that addresses all three deficiencies simultaneously. SLATE combines: (i) a spectrally-constrained compression operator providing 1-Lipschitz behaviour, bounded gradients, and information-monotonicity by the data processing inequality; (ii) a Fisher-information weighted token selection rule providing improved sample complexity over cosine top-k under standard regularity conditions; and (iii) a Lyapunov-derived annealing schedule with explicit bifurcation guard ensuring the sparse-to-dense transition does not pass through curvature-degenerate regions.

2. Related Work

Sparse and compressed attention. Linformer (Wang et al., 2020) projects keys and values onto a low-dimensional space. Performer (Choromanski et al., 2020) uses random-feature kernel approximations. Longformer and BigBird use predefined sparse patterns. FlashAttention (Dao et al., 2022) computes exact attention more efficiently via tiling. NSA (Yuan et al., 2025) introduces native-trainable sparse attention with substantial speedups at long contexts. SLATE builds directly on NSA; all NSA properties are inherited.

Spectral normalisation. Miyato et al. (2018) introduced spectral normalisation for GAN discriminators, constraining weight matrices to have $\sigma_1 \leq 1$ and providing 1-Lipschitz behaviour. The technique has subsequently been applied to a variety of architectures for training stability. To our knowledge, no published work applies spectral normalisation specifically to the compression operator of a compressed-attention method, nor combines it with the data processing inequality argument to obtain information-monotonicity of the compression step.

Information-geometric optimisation. Natural gradient methods (Amari, 1998), K-FAC (Martens and Grosse, 2015), and Shampoo (Gupta et al., 2018) precondition gradient updates by Fisher information matrix or its approximations. To our knowledge, no published work applies the Fisher information criterion to token selection in compressed attention.

Lyapunov stability theory. Lyapunov methods have been applied to the analysis of gradient flow optimisation but, to our knowledge, have not been used to derive annealing schedules for sparse-to-dense attention transitions.

3. The SLATE Framework

3.1 Spectrally-constrained compression operator (Innovation 3)

We constrain the compression operator G to satisfy $\sigma_1(G) \leq 1$. Two enforcement regimes are provided:

Approximate enforcement via spectral normalisation: $G = W / \max(1, \hat{\sigma}_1(W))$, where $\hat{\sigma}_1(W)$ is a power-iteration estimate of the largest singular value of W . Maximum singular values typically lie

in the range 1.01 to 1.05.

Exact enforcement via singular value decomposition projection: after each optimiser step, compute $W = U \text{diag}(s) V^T$, clamp $s_i \leftarrow \min(s_i, 1)$, and reconstruct $W \leftarrow U \text{diag}(s_{\text{clamped}}) V^T$. Maximum singular values lie within machine precision of unity (we measure 1.000079 across 16 layers).

The exact-enforcement regime is required for formal verification and safety-critical deployments. The approximate regime is computationally cheaper and is appropriate for most research settings.

Theorem 1 (Information monotonicity). Under the constraint $\sigma_1(G) \leq 1$, the compression operator G satisfies the data processing inequality:

$$H(G(X)) \leq H(X)$$

where H denotes differential entropy and X is the input tensor distribution.

Proof sketch. A 1-Lipschitz mapping cannot increase the variance of its input distribution along any direction. The differential entropy of a distribution is bounded above by the entropy of a Gaussian with matching covariance, and the Gaussian entropy is monotonically non-decreasing in variance per dimension. The data processing inequality applied to the Markov chain $X \rightarrow G(X) \rightarrow Y$ completes the argument.

3.2 Fisher-information weighted token selection (Innovation 2)

In place of cosine top-k selection (the standard rule in NSA and related methods), SLATE specifies a Fisher utility:

$$\phi_{i\alpha} = \nabla L_{i\alpha}^T F^{-1} \nabla L_{i\alpha}$$

where $\nabla L_{i\alpha}$ is the loss gradient contribution from the (i, α) attention pair and F is the Fisher information matrix. The diagonal Fisher $F_{jj} = E[(\partial_{\theta_j} L)^2]$ may be reused from the second-moment estimate v_t maintained by Adam-class optimisers, incurring no additional computational cost beyond gradient computation. Selection is then $S_i = \text{top-}k_{\alpha} \{ \phi_{i\alpha} \}$.

Under the isotropic Fisher special case $F \propto I$, the Fisher utility reduces exactly to inner-product top-k, recovering the cosine selection rule of NSA as a degenerate case.

Note on experimental validation. Innovation 2 is specified theoretically in this paper and in the underlying patent application. The implementation required for proper experimental validation involves modification of the NSA internal block-selection logic, which is deferred to future work.

3.3 Lyapunov-derived annealing schedule (Innovation 1)

The schedule parameter $\beta(t) \in [0, 1]$ controls the convex combination $L(t) = \beta(t) L_{\text{CSA}} + (1 - \beta(t)) L_{\text{full}}$ of the compressed-selective attention loss and the full attention loss. SLATE specifies the Lyapunov-derived rate:

$$d\beta/dt = -\epsilon \lambda_{\min}(H(t))^2 / \|\Delta(t)\|$$

where $\lambda_{\min}(H(t))$ is the smallest eigenvalue of the loss Hessian, $\|\Delta(t)\|$ is the norm of the gradient mismatch between compressed-selective and full attention loss components, and ϵ is a stability tolerance.

The rate is derived from the requirement that the Lyapunov function $V(e) = \|\theta(t) - \theta^*(t)\|^2 / 2$ satisfy $dV/dt \leq 0$ outside a contraction ball, where $\theta^*(t)$ is the instantaneous minimiser of the time-varying loss.

Bifurcation guard. The schedule is suspended whenever $\lambda_{\min}(H(t))$ falls below a curvature threshold η : $d\beta/dt = 0$. This guard ensures the schedule cannot advance through curvature-degenerate regions where the linearised stability analysis breaks down.

$\lambda_{\min}(H)$ is estimated online via 3-5 Lanczos iterations on stochastic Hessian-vector products computed via the Pearlmutter trick over a sub-batch.

4. Experimental Validation

4.1 Setup

We validate SLATE at toy scale on character-level Tiny Shakespeare with a 1.85M-parameter transformer (4 layers, hidden 192, 6 heads), sequence length 256, batch 32, 1500 training steps, AdamW with learning rate $3e-4$. The NSA backbone is the lucidrains implementation. Compression layers are `k_compress` and `v_compress` modules across all 4 attention blocks (16 layers total).

4.2 Spectral constraint validation

We compare four conditions: C1 (full attention baseline), C2 (NSA baseline), C3-strict (SLATE Innovation 3 with SVD projection), and C6 (SLATE-partial, Innovations 1+3 combined).

Condition	Val Loss	Perplexity	Max σ_1	Violations	Max ∇
C1 Full	1.9036	6.71	n/a	n/a	1.18
C2 NSA	1.6800	5.37	n/a	n/a	1.22
C3-strict	1.6814	5.37	1.000079	0/16	1.22
C6 SLATE	1.7137	5.55	1.000085	0/16	1.18

Innovation 3 is fully validated: quality is preserved within 0.0014 of NSA baseline, the spectral constraint holds to within machine precision across all 16 compression layers throughout training, and gradients are bounded.

4.3 Information monotonicity validation (Theorem 1)

On the C3-strict model architecture with spectral projection applied, we test the data processing inequality $H(G(X)) \leq H(X)$ using two independent entropy estimators applied to inputs and outputs of all 16 compression layers on a 64-sample validation batch.

Gaussian entropy estimator (per-dimension parametric):

- Mean delta H (output - input) = -0.55 nats per dimension
- Violations: 0 / 16 layers

Spectral entropy estimator (non-parametric, SVD-based):

- Mean delta H (output - input) = -1.78 nats

- Violations: 0 / 16 layers

Both estimators agree across all layers. Theorem 1 is empirically supported.

4.4 Lyapunov schedule validation

We compare three schedule controllers under matched conditions: constant β , heuristic tanh schedule, and Lyapunov-derived schedule with bifurcation guard. All three train successfully. The Lyapunov schedule produces validation loss 1.7113, comparable to tanh (1.6961) and constant (1.6801). The schedule mechanism (Lanczos eigenvalue estimation, rate computation, guard branch) executes correctly. However, under toy training conditions the Lyapunov schedule did not produce measurable advantage over heuristic schedules: the bifurcation guard did not activate (no curvature degeneracy encountered) and the schedule parameter did not vary substantially from its initial value (curvature signals too small to drive updates). The schedule mechanism is validated; behavioural advantage under stressed conditions is deferred to production-scale validation.

4.5 Limitations of toy-scale validation

The validation reported here is explicitly at toy scale. Production-scale conclusions cannot be drawn from these experiments. Key limitations:

- Sequence length 256 is well below the asymptotic regime in which NSA shows speedup. At our scale, NSA itself is 4.44× slower than full attention.
- Innovation 2 (Fisher selection) is not implemented in working code due to the need for NSA-internal modifications.
- Single random seed (42) used throughout; statistical variance not estimated.
- Single dataset and single model size.

Production-scale validation (Pythia-410M, full-context, multiple seeds) is planned and will appear in an updated version of this preprint.

5. Operator-Theoretic Interpretation

The spectral constraint $\sigma_1(G) \leq 1$ places G within the operator class of sub-unitary completely-positive trace-non-increasing maps (sub-unitary CPTN) from quantum information theory. This is more than analogy: the same operator-norm bound that protects training stability in the classical neural-network regime also protects information-theoretic security guarantees in the quantum regime, and the same data-processing-inequality argument applies in both.

This identification has practical consequences. SLATE-trained models are amenable to the same formal verification techniques developed for quantum circuits, in which operator-norm bounds are sufficient to certify input-output behaviour. For deployment in safety-critical, regulated, or formally-verified AI contexts, this is the operative property: not just training stability but a foundation for mathematical certification of behaviour.

6. Discussion

SLATE is positioned as a contribution to verified and safety-critical AI training rather than as a direct speedup over NSA. SLATE inherits NSA's computational efficiency without claiming to

improve on it. The contribution is in mathematical guarantees: provable stability, provable information monotonicity, and bounded gradient propagation, all available simultaneously in a single training framework.

Three audiences may find SLATE useful. First, AI safety researchers and formal verification teams who need provable bounds on training-time behaviour. Second, deployment teams in regulated industries (medical, defence, financial) where mathematical certification supplements empirical testing. Third, hardware accelerator designers who can exploit bounded-gradient training to reduce safety margins in chip design.

7. Conclusion

We have introduced SLATE, a training framework extending NSA with three theoretically-motivated innovations providing provable stability guarantees not available in published sparse-attention methods. The spectral-constraint innovation and its information-monotonicity consequence are fully validated at toy scale. The combined system of Innovations 1+3 trains stably and preserves quality. Fisher selection and Lyapunov-schedule advantage are specified theoretically with experimental validation deferred.

Code, reproducibility notebooks, and experimental results are available at <https://github.com/physivitis/slate> under the MIT licence. The framework is the subject of UK Patent Application GB2612127.7.

References

- Amari, S. (1998). Natural gradient works efficiently in learning. *Neural Computation*.
- Choromanski, K., et al. (2020). Rethinking attention with Performers. *ICLR 2021*.
- Dao, T., et al. (2022). FlashAttention. *NeurIPS 2022*.
- Gupta, V., Koren, T., Singer, Y. (2018). Shampoo. *ICML 2018*.
- Martens, J., Grosse, R. (2015). K-FAC. *ICML 2015*.
- Miyato, T., et al. (2018). Spectral normalisation for GANs. *ICLR 2018*.
- Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS 2017*.
- Wang, S., et al. (2020). Linformer. arXiv:2006.04768.
- Yuan, J., et al. (2025). Native Sparse Attention. *ACL 2025*.

Acknowledgements

The author thanks the DeepSeek team for the open-source release of NSA, on which this work directly builds. The reference NSA implementation used is from Phil Wang's lucidrains/native-sparse-attention-pytorch package.